

Week 1 Lab

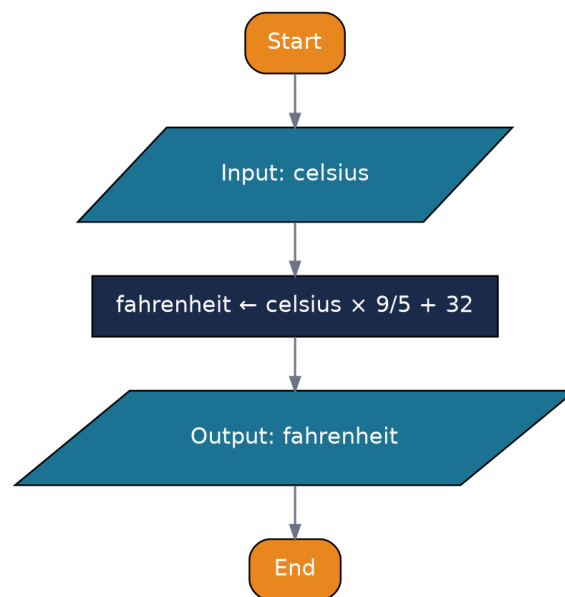
Flowchart Practice: Sequence, Decision, Loop

DELIVERED BY:	Teaching Assistants
FOLLOWS:	Lecture — From Algorithms to Flowcharts
FORMAT:	Pen and paper, or a diagramming tool — no Java code in this lab

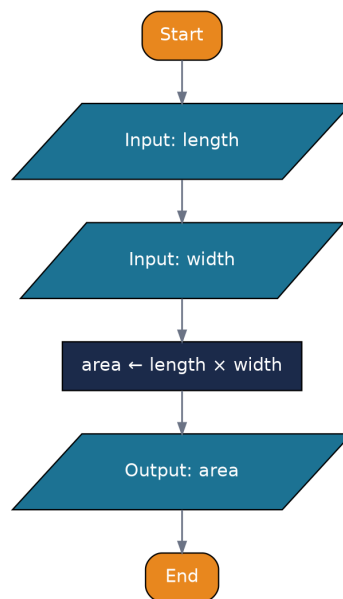
GROUND RULES

- Use only the four symbols from lecture: Terminal (Start/End), Process, Decision, Input/Output.
- Every flowchart needs exactly one Start and at least one End.
- Every Decision diamond needs exactly two labeled exits: Yes and No.
- Each question's flowchart, pseudocode, and (for loops) trace table are all on one page, right after the question. Cover them while solving.

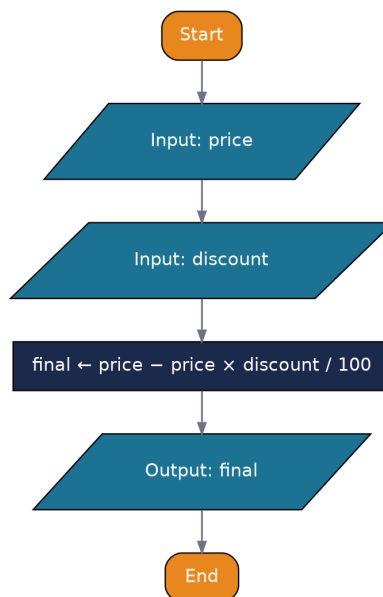
Q1. Input a temperature in Celsius, and output the equivalent in Fahrenheit.

FLOWCHART**PSEUDOCODE**

1. input celsius
2. fahrenheit $\leftarrow$ celsius $\times$ 9/5 + 32
3. output fahrenheit

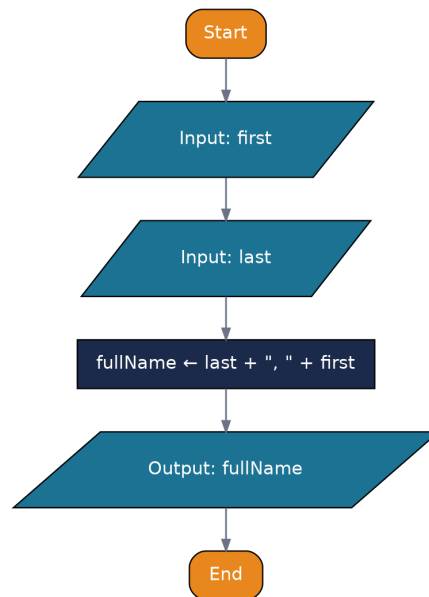
Q2. Input the length and width of a rectangle, and output its area.**FLOWCHART****PSEUDOCODE**

1. input length
2. input width
3. $\text{area} \leftarrow \text{length} \times \text{width}$
4. output area

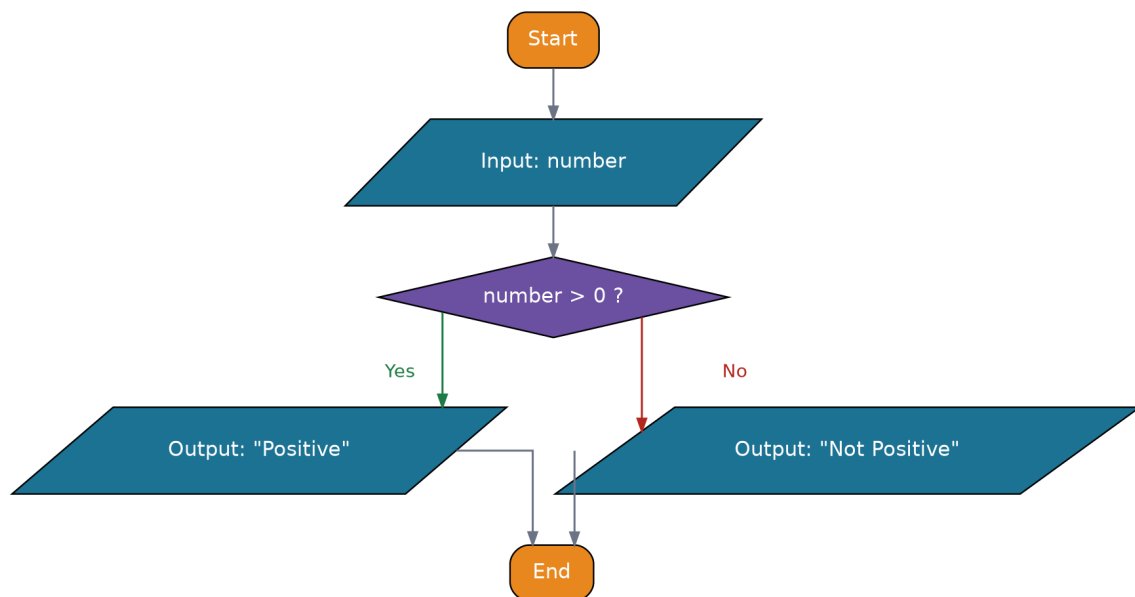
Q3. Input a price and a discount percentage, and output the final price.**FLOWCHART****PSEUDOCODE**

1. input price
2. input discount
3. $\text{final} \leftarrow \text{price} - \text{price} \times \text{discount} / 100$
4. output final

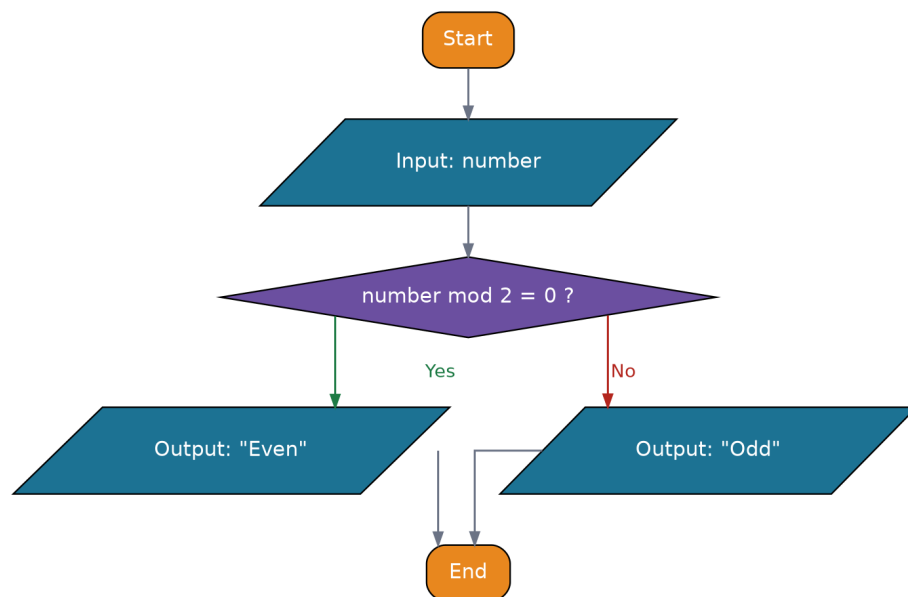
Q4. Input someone's first name and last name, and output them combined as "Last, First".

FLOWCHART**PSEUDOCODE**

1. input first
2. input last
3. fullName ← last + ", " + first
4. output fullName

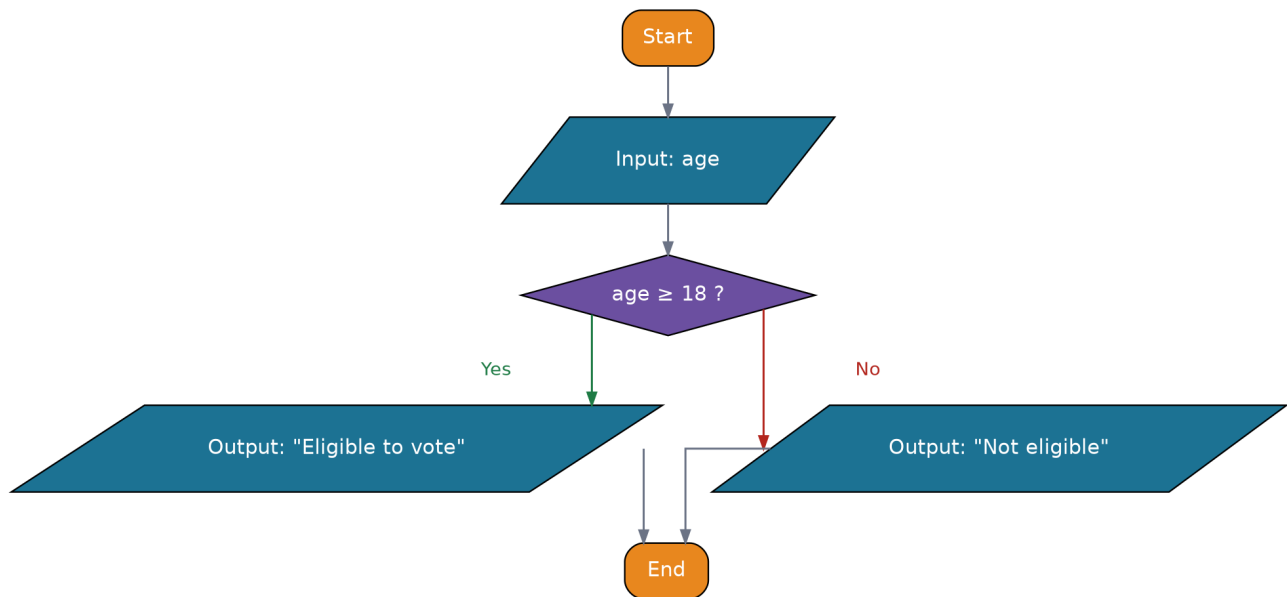
Q5. Input a number; output “Positive” or “Not Positive.”**FLOWCHART****PSEUDOCODE**

```
1. input number
2. if number > 0 then
3.   output "Positive"
4. else
5.   output "Not Positive"
6. end if
```

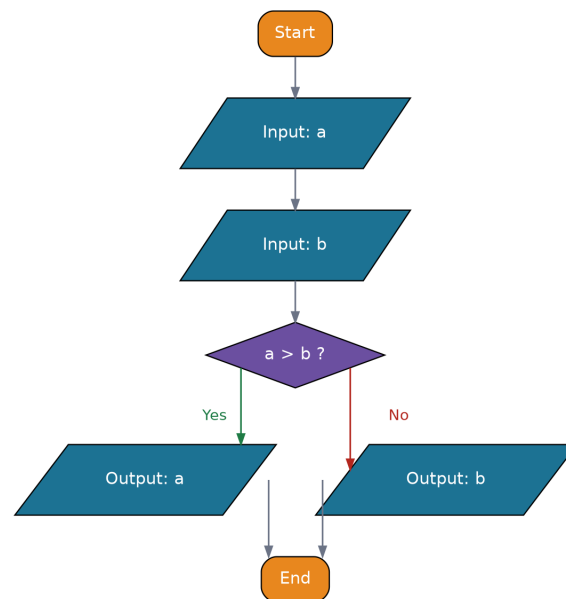
Q6. Input a number; output “Even” or “Odd.”**FLOWCHART****PSEUDOCODE**

```
1. input number
2. if number mod 2 = 0 then
3.   output "Even"
4. else
5.   output "Odd"
6. end if
```

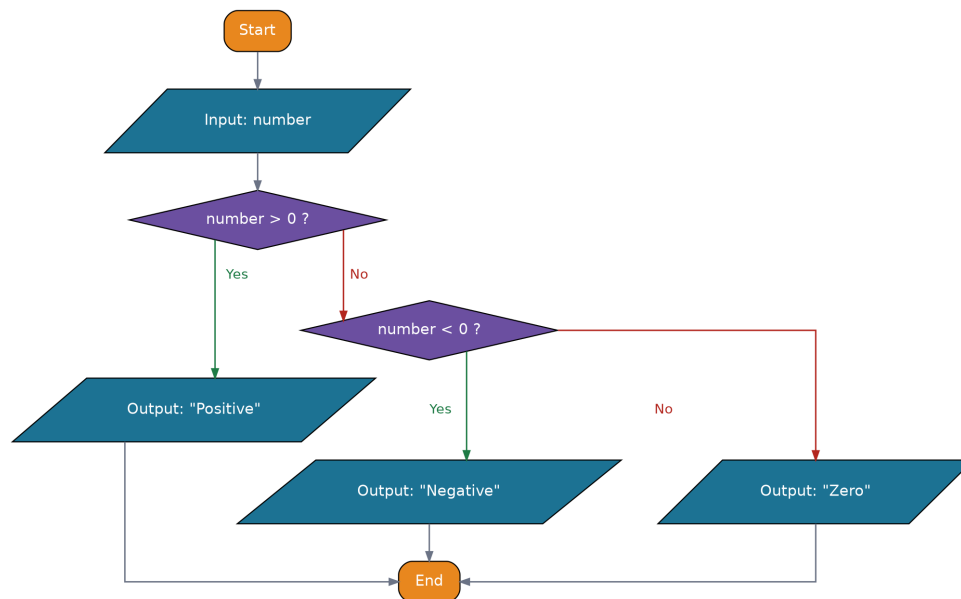
Q7. Input an age; output “Eligible to vote” if 18 or older, otherwise “Not eligible.”

FLOWCHART**PSEUDOCODE**

```
1. input age
2. if age ≥ 18 then
3.   output "Eligible to vote"
4. else
5.   output "Not eligible"
6. end if
```

Q8. Input two numbers; output the larger of the two.**FLOWCHART****PSEUDOCODE**

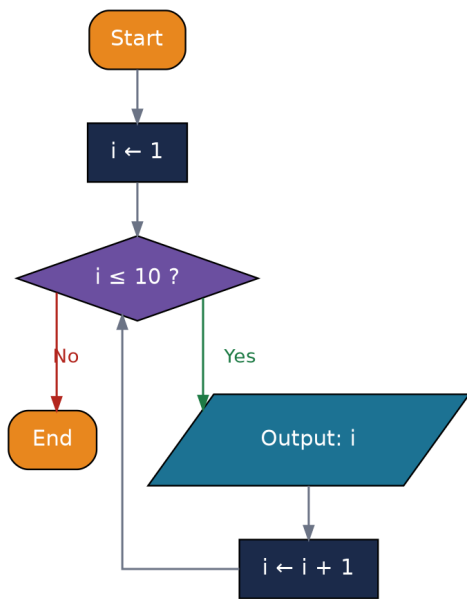
```
1. input a
2. input b
3. if a > b then
4.     output a
5. else
6.     output b
7. end if
```

Q9. Input a number; output “Positive,” “Negative,” or “Zero.”**FLOWCHART****PSEUDOCODE**

```
1. input number
2. if number > 0 then
3.   output "Positive"
4. else if number < 0 then
5.   output "Negative"
6. else
7.   output "Zero"
8. end if
```

Q10. Print the numbers 1 through 10.

FLOWCHART



PSEUDOCODE

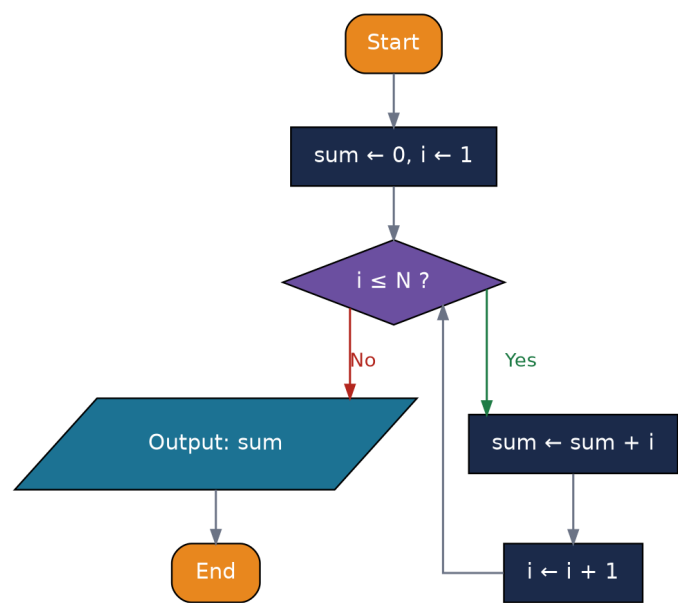
```
1. i ← 1
2. while i ≤ 10 do
3.   output i
4.   i ← i + 1
5. end while
```

TRACE TABLE

i	i ≤ 10 ?	output
1	Yes	1
2	Yes	2
3	Yes	3
...	...	...
10	Yes	10
11	No	(exit)

Q11. Input a number N; compute and output the sum $1 + 2 + \dots + N$.

FLOWCHART



PSEUDOCODE

```
1. input N
2. sum ← 0, i ← 1
3. while i ≤ N do
4.     sum ← sum + i
5.     i ← i + 1
6. end while
7. output sum
```

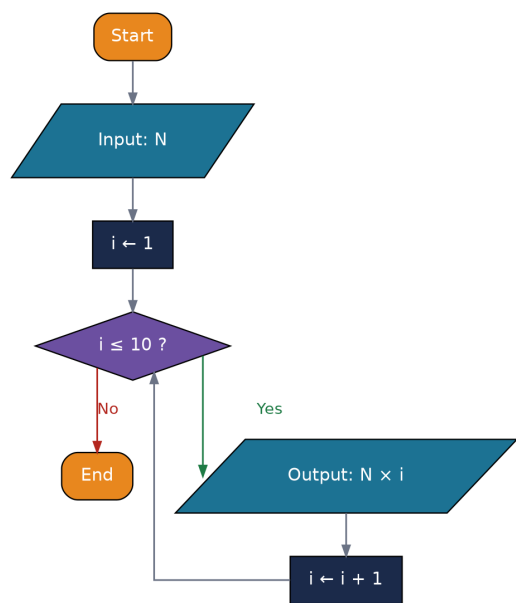
TRACE TABLE

Sample run with N = 4.

i	i ≤ 4 ?	sum ← sum+i
1	Yes	0+1=1
2	Yes	1+2=3
3	Yes	3+3=6
4	Yes	6+4=10
5	No	(exit – output 10)

Q12. Input a number N; print the multiplication table of N (N×1 through N×10).

FLOWCHART



PSEUDOCODE

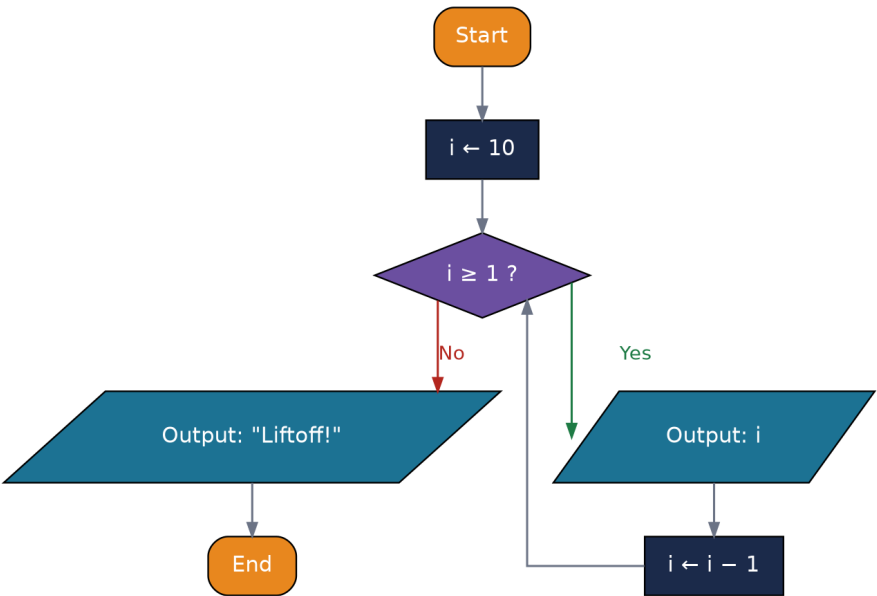
```
1. input N
2. i ← 1
3. while i ≤ 10 do
4.   output N × i
5.   i ← i + 1
6. end while
```

TRACE TABLE

i	i ≤ 10 ?	output (N=3)
1	Yes	3
2	Yes	6
3	Yes	9
...	...	...
10	Yes	30
11	No	(exit)

Q13. Count down from 10 to 1, then output “Liftoff!”

FLOWCHART



PSEUDOCODE

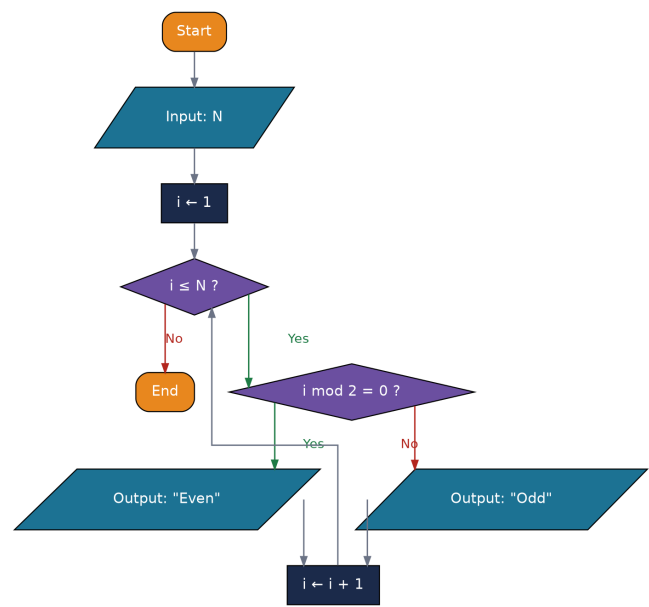
```
1. i ← 10
2. while i ≥ 1 do
3.   output i
4.   i ← i - 1
5. end while
6. output "Liftoff!"
```

TRACE TABLE

i	i ≥ 1 ?	output
10	Yes	10
9	Yes	9
...	...	...
1	Yes	1
0	No	(exit – "Liftoff!")

Q14. Input a number N; loop from 1 to N, and for each number output whether it is even or odd.

FLOWCHART



PSEUDOCODE

```
1. input N
2. i ← 1
3. while i ≤ N do
4.   if i mod 2 = 0 then
5.     output "Even"
6.   else
7.     output "Odd"
8.   end if
9.   i ← i + 1
10. end while
```

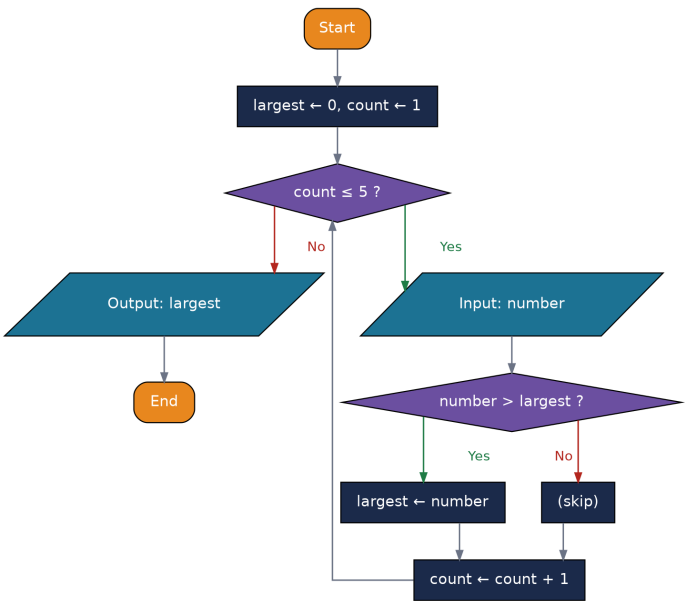
TRACE TABLE

Sample run with N = 4.

i	i ≤ 4 ?	i mod2=0?	output
1	Yes	No	Odd
2	Yes	Yes	Even
3	Yes	No	Odd
4	Yes	Yes	Even
5	No	—	(exit)

Q15. Input 5 numbers one at a time; output the largest one at the end.

FLOWCHART



PSEUDOCODE

```
1. largest ← 0, count ← 1
2. while count ≤ 5 do
3.   input number
4.   if number > largest then
5.     largest ← number
6.   end if
7.   count ← count + 1
8. end while
9. output largest
```

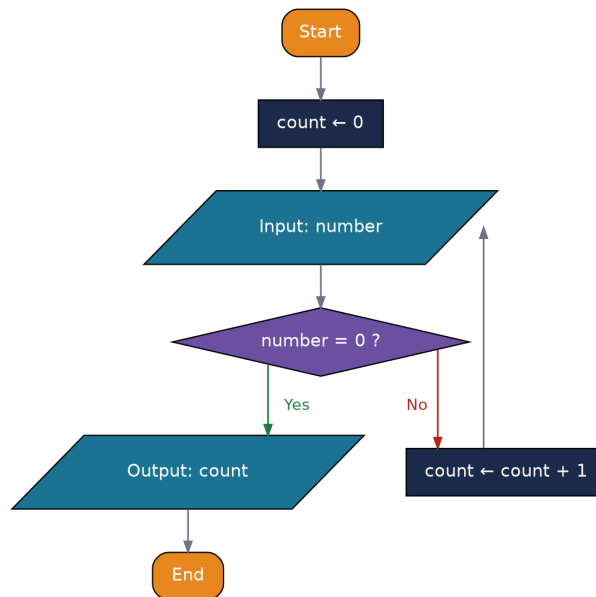
TRACE TABLE

Sample run with inputs 3, 7, 2, 9, 5.

count	number	num>largest?	largest
1	3	Yes	3
2	7	Yes	7
3	2	No	7
4	9	Yes	9
5	5	No	9
6 (exit)	—	—	output 9

Q16. Keep asking the user to input a number until they enter 0; then output how many numbers they entered (not counting the 0).

FLOWCHART



PSEUDOCODE

```

1. count ← 0
2. input number
3. while number ≠ 0 do
4.   count ← count + 1
5.   input number
6. end while
7. output count
  
```

Loop-back returns to *Input: number*, not the decision — input happens again every pass.

TRACE TABLE

Sample run with inputs 5, 3, 8, 0.

number entered	count
5	1
3	2
8	3
0 (stop)	output 3

WRAP-UP & INSTRUCTOR NOTES

Wrap-up check (5 min)

Before leaving, each student should be able to point to their Part D or Part E flowchart and answer out loud:

- “Where does my loop check its condition?”
- “What happens the moment the condition becomes false?”

Instructor notes

- Do not let students write Java for these — the point is separating logic from syntax. Redirect anyone who starts typing code back to the diagram.
- Each question page has three views side by side: flowchart, pseudocode, and (for loops) a trace table. Encourage students to solve with the flowchart first, then check themselves against the pseudocode and trace.
- Common mistakes to watch for: a decision with only one exit arrow, a loop with no way to ever become false (infinite loop on paper), forgetting to update the loop's counter/condition variable inside the loop body, and — in Q16 — looping back to the decision instead of back to the input step.
- Q9 and Part E are intentionally harder — pace the room so most students reach at least the start of Part D; Part E can be finished as homework.
- The diagrams, pseudocode, and traces in this packet show one correct shape, not the only one — a student's version with a different but logically equivalent order is still correct.
- Next lecture: translating these exact flowcharts into Java if/else and while/for syntax — keep today's diagrams, students will reuse them.